

Shell

PENG

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1 Introduction

<https://www.gnu.org/software/bash/manual/bash.html>

2 Shell Syntax

2.1 Quoting

Definition 2.1 (quoting). *Quoting is used to remove the special meaning of certain characters or words to the shell.*

```
echo "${var}"
echo "$(date)"
echo "`date`"
echo -e "\t"
# \n : newline
# \t : horizontal tab
# \r : carriage return
```

2.2 Pipelines

Definition 2.2 (pipeline). *cmd1 output $\rightarrow$ cmd2 input*

```
cmd1 | cmd2      # stdout
cmd1 |& cmd2      # stderr to stdout
cmd1 2>&1 | cmd2  # stderr to stdout
```

2.3 Lists of Commands

Definition 2.3 (list).

```
cmd1 && cmd2 # And, cmd1 true then cmd2
cmd1 || cmd2 # Or, cmd1 false then cmd2
cmd1; cmd2
cmd1 & cmd2 # cmd1 async
```

2.4 Construct

2.4.1 Looping

Definition 2.4 (looping).

```
for item in item1 item2 ...; do
    cmd
done
```

2.4.2 Conditional

Definition 2.5 (conditional).

```
if [[ cond1 ]]; then
    cmd1
elif [[ cond2 ]]; then
    cmd2
else
    cmd3
fi
```

```
# cond
# ==
# !=
# =~ : unquoted regex
```

2.4.3 Grouping

Definition 2.6 (grouping). (expr) # subshell env, var vanish when exit

3 AWK

<https://www.gnu.org/software/gawk/manual/gawk.html>

Definition 3.1 (awk). *The basic function of awk is to search files for lines that contain certain patterns.*